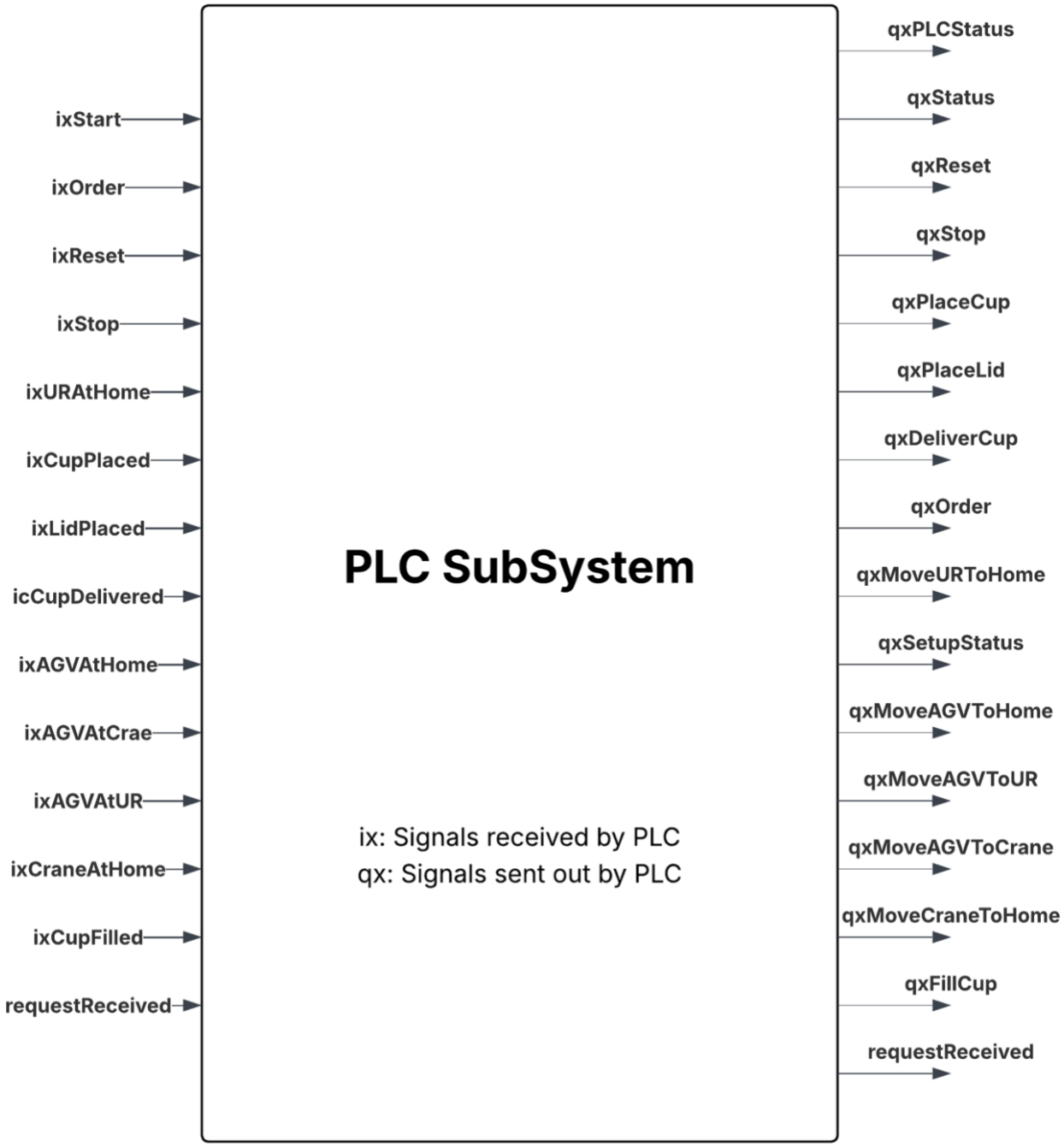
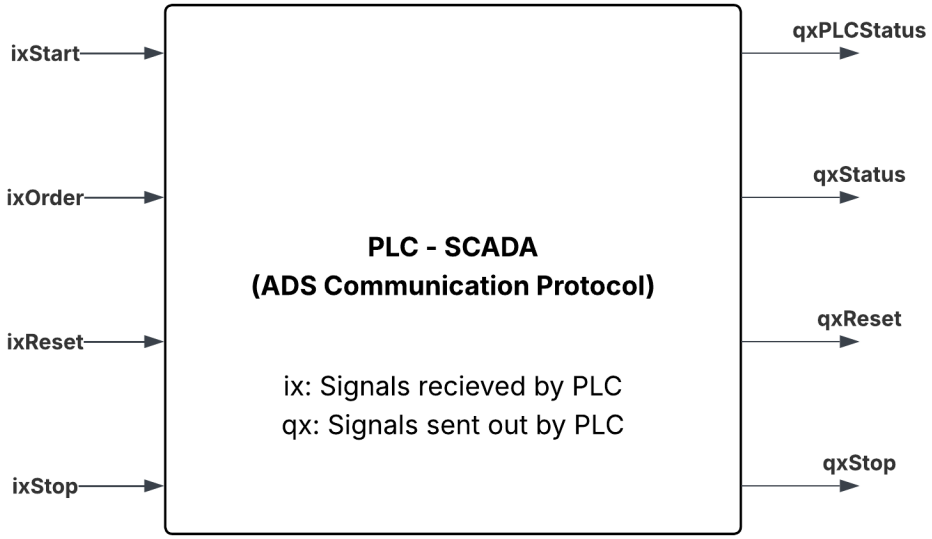
Communication between PLC and Other Systems



PLC – SCADA Communication (ADS protocol)



Input Signals from SCADA to PLC:

ixOrder (BOOL): The signal tells PLC the order ID. The order ID is either 1 (Small Cup) or 2 (Large Cup).

ixStart (BOOL): The signal tells PLC to start operation. The order ID must be sent to PLC first before sending this signal to the PLC.

ixStop (BOOL): The signal tells PLC to stop operation. Once operation is stopped, *qxStop* becomes True.

ixReset (BOOL): The signal tells PLC to reset all systems and start operation from start. The operation must have been stopped (*qxStop: True*) before *ixReset* signal should be sent to PLC.

Output Signals from PLC to SCADA:

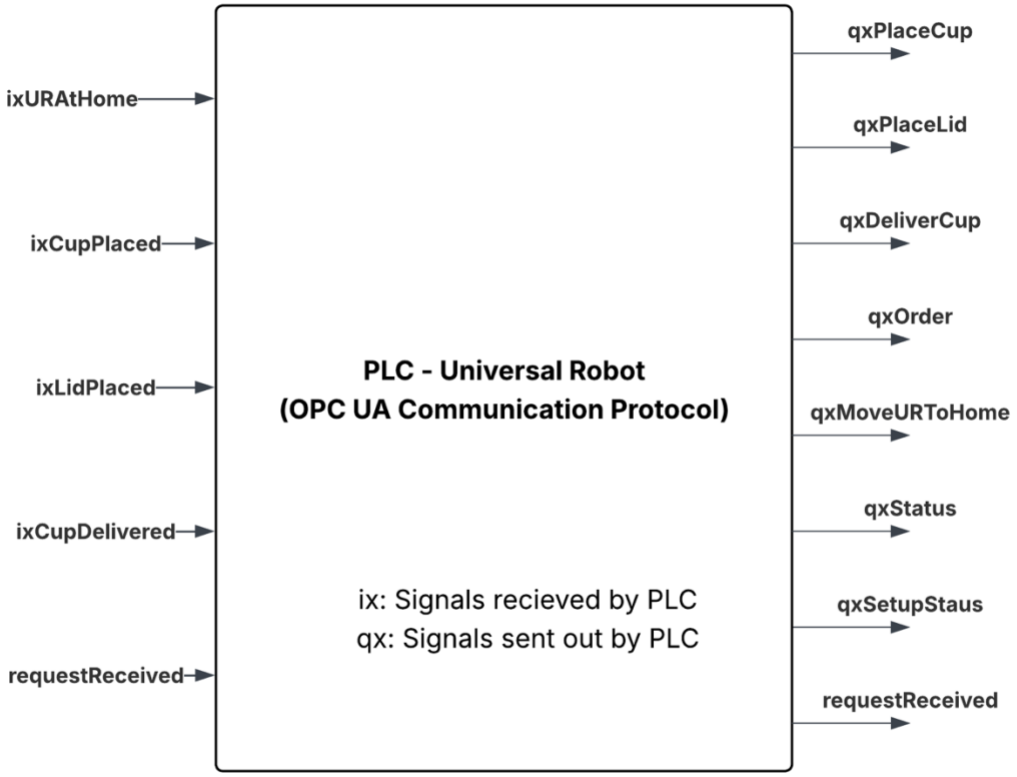
qxPLCStatus (STRING): The status of operation e.g. “Ready to start operation”

qxStatus (INT): The status code of operation e.g. 10

qxReset (BOOL): *True* when Operation is in Resetting state.

qxStop (BOOL): *True* when operation is stop state. *False* if operation is ongoing.

PLC – UR Communication (OPC UA)



Input Signals from UR to PLC:

ixURAtHome (BOOL): The signal tells PLC that Universal Robot has arrived at home position.

ixCupPlaced (BOOL): The signal tells PLC that Universal Robot has placed cup on the AGV.

ixLidPlaced (BOOL): The signal tells PLC that Universal Robot has placed lid on the cup.

ixCupDelivered (BOOL): The signal tells PLC that Universal Robot has delivered the cup.

requestReceived (BOOL): Handshake that confirms that Universal Robot received a signal from PLC. E.g. if PLC instructs UR to move to home position (*qxMoveURToHome*: *True*), UR confirms (*requestReceived :True*) to PLC that the signal is received.

Output Signals from PLC to UR:

qxPlaceCup (BOOL): Tells UR to place cup on the AGV.

qxPlaceLid (BOOL): Tells UR to place lid on the cup.

qxDeliverCup (BOOL): Tells UR to deliver cup.

qxMoveURToHome (BOOL): Tells UR to go to home position.

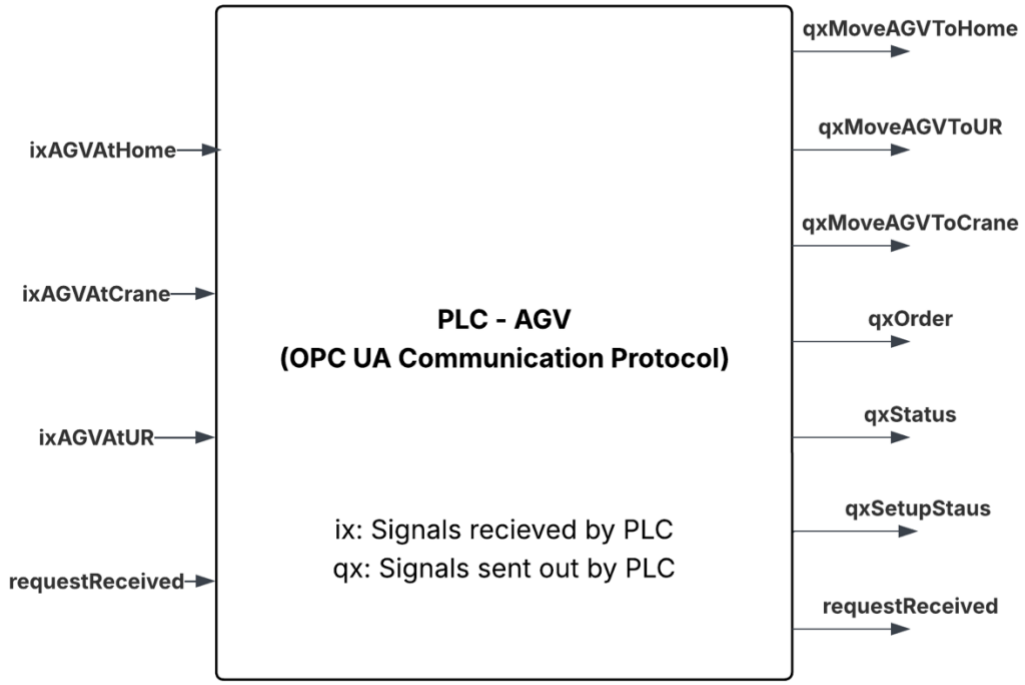
qxOrder (INT): The Order ID of the selected cup. The order ID is either 1 (Small Cup) or 2 (Large Cup).

qxStatus (INT): The status code of operation e.g. 10

qxSetupStatus (INT): The status code of the setup operation e.g. 1100

requestReceived (BOOL): Handshake that confirms that PLC received a signal from UR. E.g. if UR has arrived at home position (*ixURAtHome*: *True*), PLC confirms (*requestReceived :True*) to UR that the signal is received.

PLC – AGV Communication (OPC UA)



Input Signals from AGV to PLC:

ixAGVAtHome (BOOL): The signal tells PLC that AGV has arrived at home position.

ixAGVAtCrane (BOOL): The signal tells PLC that AGV has arrived at crane position.

ixAGVAtUR (BOOL): The signal tells PLC that AGV has arrived at universal robot position.

requestReceived (BOOL): Handshake that confirms that AGV received a signal from PLC. E.g. if PLC instructs AGV to move to home position (*qxMoveAGVToHome*: *True*), AGV confirms (*requestReceived :True*) to PLC that the signal is received.

Output Signals from PLC to AGV:

qxMoveAGVToHome (BOOL): Tells AGV to go to home position.

qxMoveAGVToUR (BOOL): Tells AGV to go to UR position.

qxMoveAGVToCrane (BOOL): Tells AGV to go to Crane position.

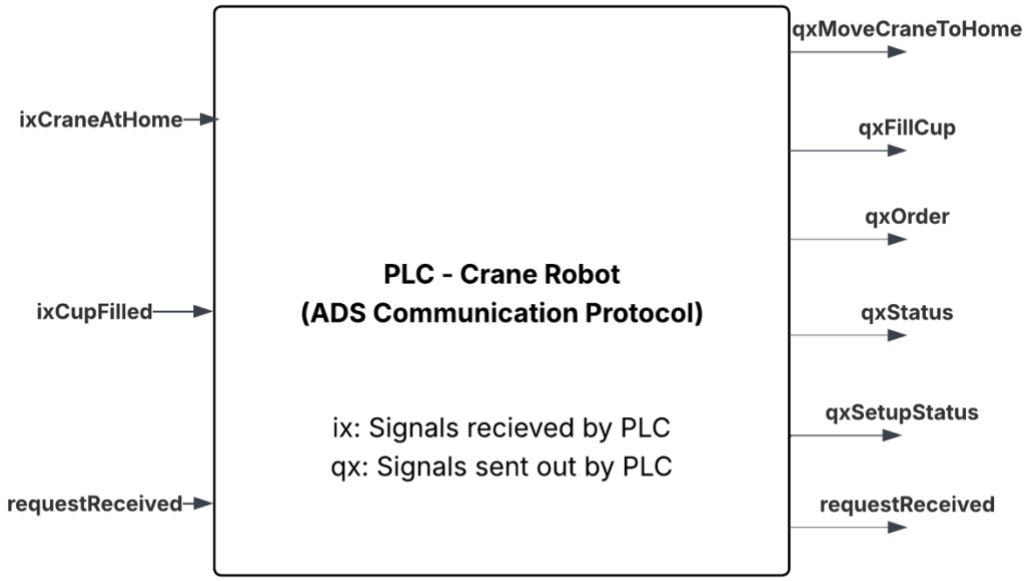
qxOrder (INT): The Order ID of the selected cup. The order ID is either 1 (Small Cup) or 2 (Large Cup).

qxStatus (INT): The status code of operation e.g. 10

qxSetupStatus (INT): The status code of the setup operation e.g. 1100

requestReceived (BOOL): Handshake that confirms that PLC received a signal from AGV. E.g. if AGV has arrived at home position (*ixAGVAtHome*: *True*), PLC confirms (*requestReceived: True*) to AGV that the signal is received.

PLC – Crane Communication (ADS Protocol)



Input Signals from Crane to PLC:

ixCraneAtHome (BOOL): The signal tells PLC that Crane is at home position.

ixCupFilled (BOOL): The signal tells PLC that Crane has filled the cup.

requestReceived (BOOL): Handshake that confirms that Crane received a signal from PLC. E.g. if PLC instructs Crane to move to home position (*qxMoveCraneToHome*: *True*), Crane confirms (*requestReceived :True*) to PLC that the signal is received.

Output Signals from PLC to Crane:

qxMoveCraneToHome (BOOL): Tells Crane to move to home position.

qxFillCup (BOOL): Tells Crane to fill the cup based on the selected Order ID.

qxOrder (INT): The Order ID of the selected cup. The order ID is either 1 (Small Cup) or 2 (Large Cup).

qxStatus (INT): The status code of operation e.g. 10

qxSetupStatus (INT): The status code of the setup operation e.g. 1100

requestReceived (BOOL): Handshake that confirms that PLC received a signal from Crane. E.g. if Crane has moved to home position (*ixCraneAtHome*: *True*), PLC confirms (*requestReceived: True*) to Crane that the signal is received.

Operation Status Code

The table below refers to the status code of signal *qxStatus*

Status Code	Description
0	Initializing
5	Setting up all systems
10	Ready to Start Operation
20	UR place cup on AGV
30	Await Confirmation for UR Place Cup
40	UR Cup Placed on AGV
50	Move AGV to Crane
60	Await Confirmation for AGV Move to Crane
70	AGV at Crane

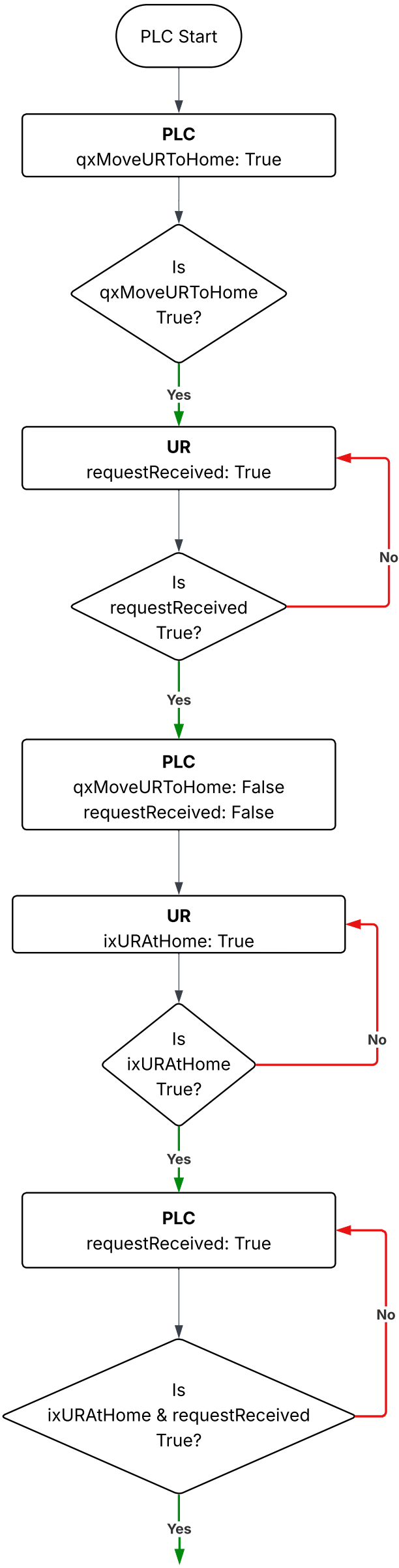
80	Crane Fill Cup
90	Await Confirmation for Crane Fill Cup
100	Cup Filled by Crane
110	Move AGV to UR
120	Await Confirmation for AGV Move to UR
130	AGV at UR
140	UR Place Lid
150	Await Confirmation for UR Place Lid
160	Lid Placed by UR
170	UR Deliver Cup
180	Await Confirmation for UR Deliver Cup
190	Cup Delivered
200	Operation Completed

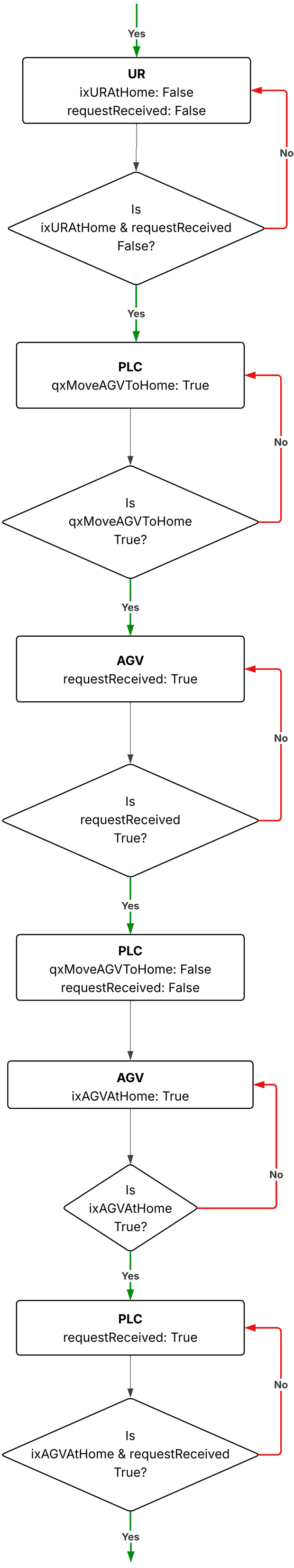
Setup Status Code

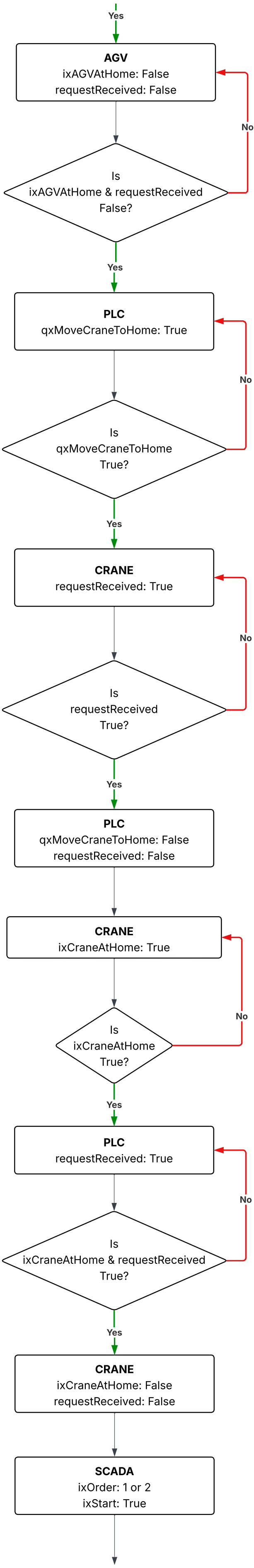
The table below refers to the status code of signal *qxSetupStatus***.** This code is only used when systems are being setup. During the setup the main operation status code (*qxStatus*) is 5

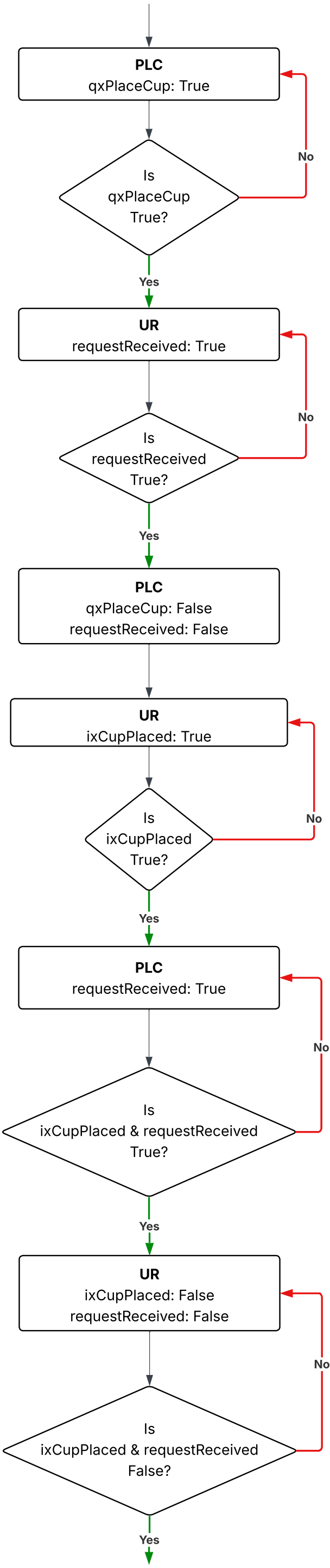
Status Code	Description
1000	Initializing
1100	Move UR to Home
1200	Await Confirmation for UR Move Home
1300	UR at Home
1400	Move AGV to Home
1500	Await Confirmation for AGV Move Home
1600	AGV at Home
1700	Move Crane to Home
1800	Await Confirmation for Crane Move Home
1900	Crane at Home

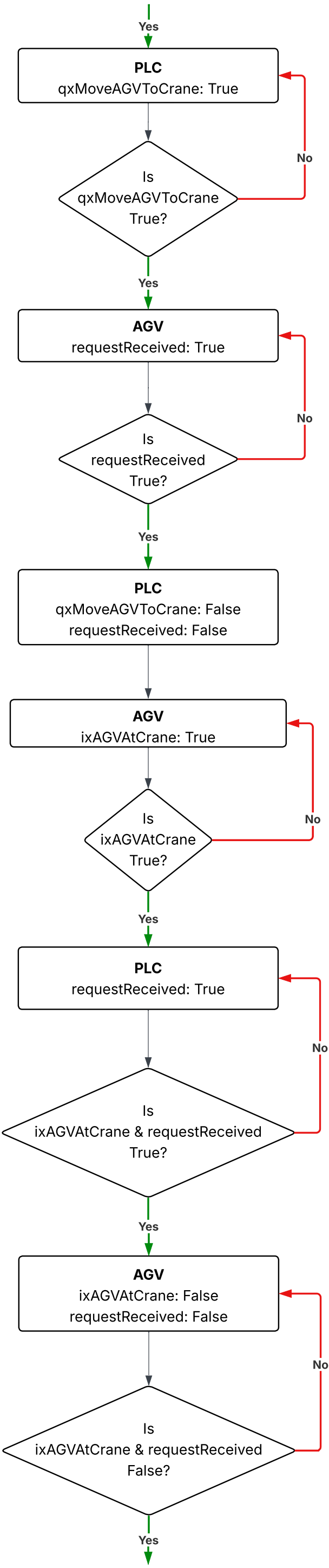
PROCESS FLOW CHARTS

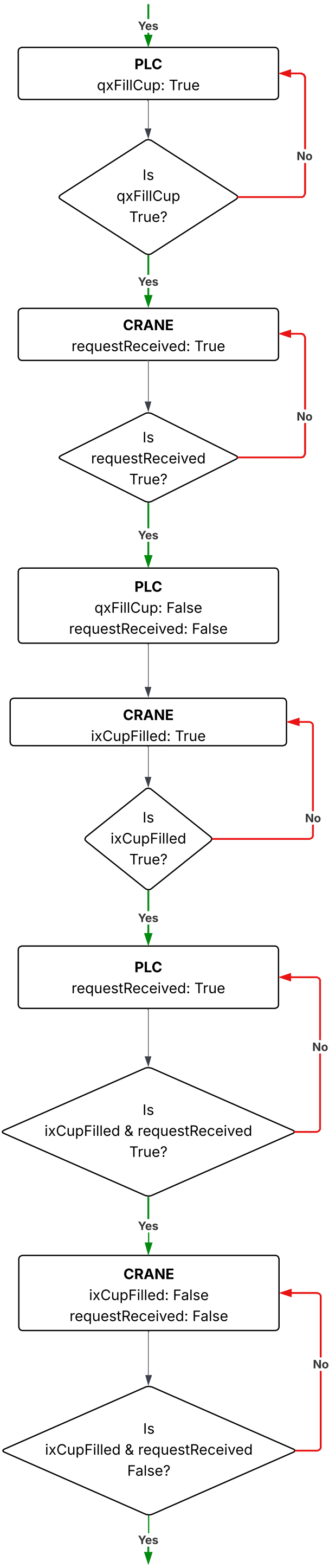

PLC Start
PLC
qxMoveURToHome: True
Is
qxMoveURToHome
True?
Yes
UR
requestReceived: True
No
Is
requestReceived
True?
Yes
PLC
qxMoveURToHome: False
requestReceived: False
UR
ixURAtHome: True
No
Is
ixURAtHome
True?
Yes
PLC
requestReceived: True
No
Is
ixURAtHome & requestReceived
True?
Yes

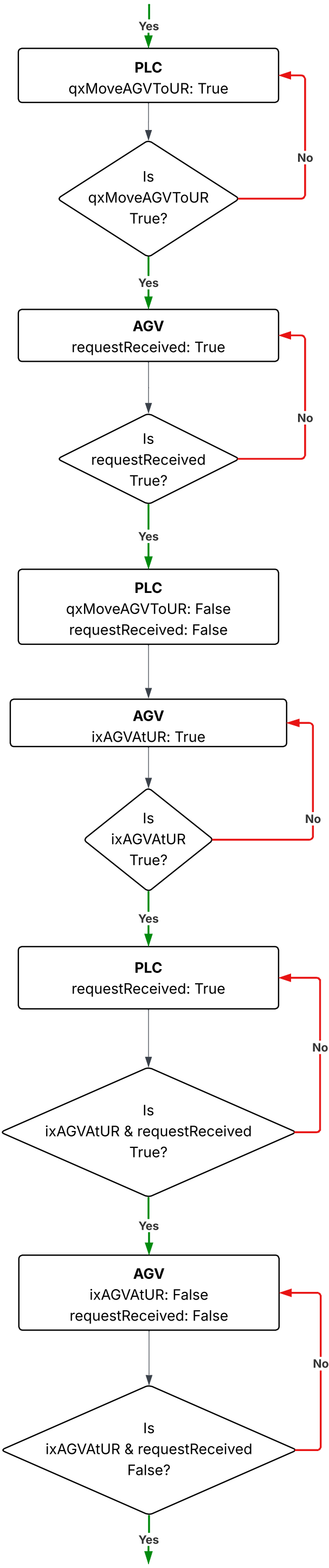

Yes
UR
ixURAtHome: False
requestReceived: False
No
Is
ixURAtHome & requestReceived
False?
Yes
PLC
qxMoveAGVToHome: True
No
Is
qxMoveAGVToHome
True?
Yes
AGV
requestReceived: True
No
Is
requestReceived
True?
Yes
PLC
qxMoveAGVToHome: False
requestReceived: False
AGV
ixAGVAtHome: True
No
Is
ixAGVAtHome
True?
Yes
PLC
requestReceived: True
No
Is
ixAGVAtHome & requestReceived
True?
Yes

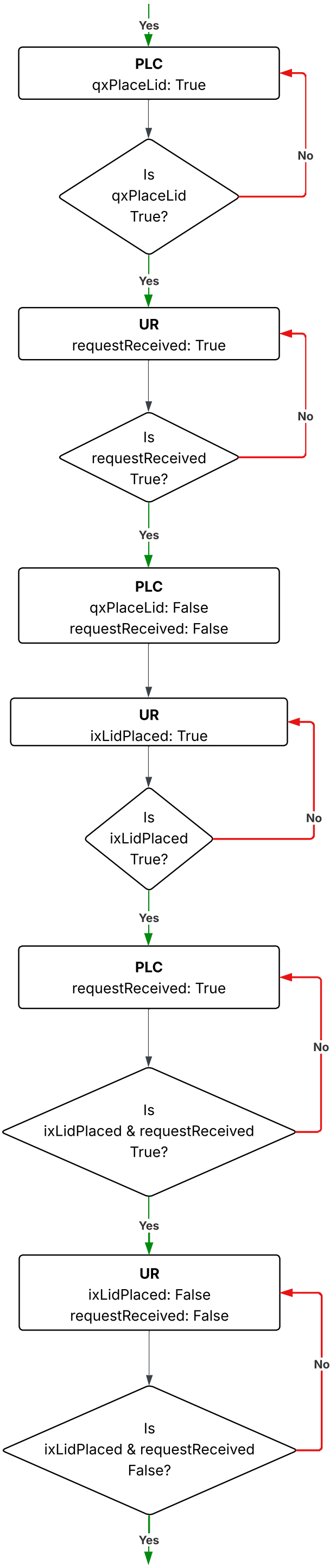

Yes
AGV
ixAGVAtHome: False
requestReceived: False
No
Is
ixAGVAtHome & requestReceived
False?
Yes
PLC
qxMoveCraneToHome: True
No
Is
qxMoveCraneToHome
True?
Yes
CRANE
requestReceived: True
No
Is
requestReceived
True?
Yes
PLC
qxMoveCraneToHome: False
requestReceived: False
CRANE
ixCraneAtHome: True
No
Is
ixCraneAtHome
True?
Yes
PLC
requestReceived: True
No
Is
ixCraneAtHome & requestReceived
True?
Yes
CRANE
ixCraneAtHome: False
requestReceived: False
SCADA
ixOrder: 1 or 2
ixStart: True

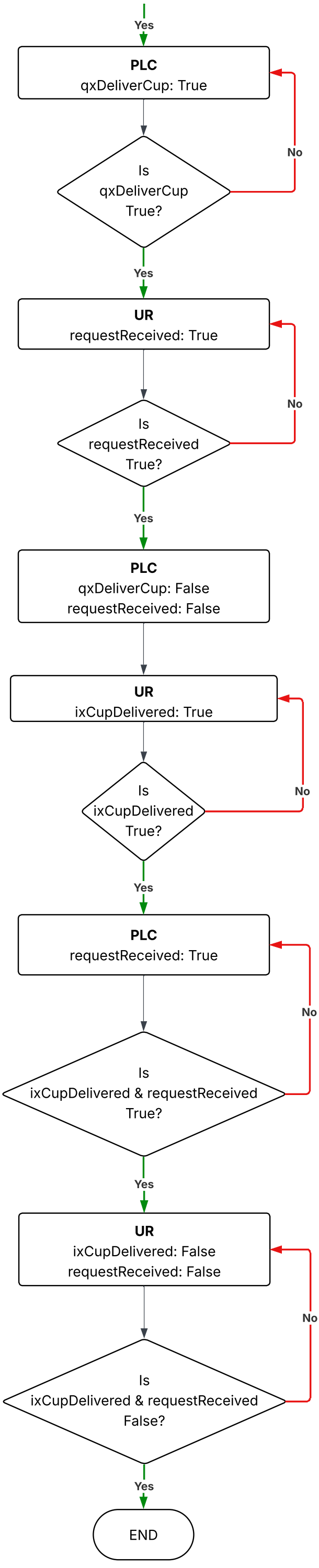

PLC
qxPlaceCup: True
Is qxPlaceCup True?
No
Yes
UR
requestReceived: True
Is requestReceived True?
No
Yes
PLC
qxPlaceCup: False
requestReceived: False
UR
ixCupPlaced: True
Is ixCupPlaced True?
No
Yes
PLC
requestReceived: True
Is ixCupPlaced & requestReceived True?
No
Yes
UR
ixCupPlaced: False
requestReceived: False
Is ixCupPlaced & requestReceived False?
No
Yes


Yes
PLC
qxMoveAGVToCrane: True
Is
qxMoveAGVToCrane
True?
No
Yes
AGV
requestReceived: True
Is
requestReceived
True?
No
Yes
PLC
qxMoveAGVToCrane: False
requestReceived: False
AGV
ixAGVAtCrane: True
Is
ixAGVAtCrane
True?
No
Yes
PLC
requestReceived: True
Is
ixAGVAtCrane & requestReceived
True?
No
Yes
AGV
ixAGVAtCrane: False
requestReceived: False
Is
ixAGVAtCrane & requestReceived
False?
No
Yes


Yes
PLC
qxFillCup: True
Is
qxFillCup
True?
No
Yes
CRANE
requestReceived: True
Is
requestReceived
True?
No
Yes
PLC
qxFillCup: False
requestReceived: False
CRANE
ixCupFilled: True
Is
ixCupFilled
True?
No
Yes
PLC
requestReceived: True
Is
ixCupFilled & requestReceived
True?
No
Yes
CRANE
ixCupFilled: False
requestReceived: False
Is
ixCupFilled & requestReceived
False?
No
Yes


Yes
PLC
qxMoveAGVToUR: True
Is
qxMoveAGVToUR
True?
No
Yes
AGV
requestReceived: True
Is
requestReceived
True?
No
Yes
PLC
qxMoveAGVToUR: False
requestReceived: False
AGV
ixAGVAtUR: True
Is
ixAGVAtUR
True?
No
Yes
PLC
requestReceived: True
Is
ixAGVAtUR & requestReceived
True?
No
Yes
AGV
ixAGVAtUR: False
requestReceived: False
Is
ixAGVAtUR & requestReceived
False?
No
Yes


Yes
PLC
qxPlaceLid: True
Is
qxPlaceLid
True?
No
Yes
UR
requestReceived: True
Is
requestReceived
True?
No
Yes
PLC
qxPlaceLid: False
requestReceived: False
UR
ixLidPlaced: True
Is
ixLidPlaced
True?
No
Yes
PLC
requestReceived: True
Is
ixLidPlaced & requestReceived
True?
No
Yes
UR
ixLidPlaced: False
requestReceived: False
Is
ixLidPlaced & requestReceived
False?
No
Yes


Yes
PLC
qxDeliverCup: True
Is
qxDeliverCup
True?
No
Yes
UR
requestReceived: True
Is
requestReceived
True?
No
Yes
PLC
qxDeliverCup: False
requestReceived: False
UR
ixCupDelivered: True
Is
ixCupDelivered
True?
No
Yes
PLC
requestReceived: True
Is
ixCupDelivered & requestReceived
True?
No
Yes
UR
ixCupDelivered: False
requestReceived: False
Is
ixCupDelivered & requestReceived
False?
No
Yes
END